

Falsifiable Predictions of the Rope Hypothesis

Four Testable Numbers and a Reproducibility Standard

Mark Palmer · June 2026

Global Networking Practice, NTT DATA, Charlotte, NC · palmer100@gmail.com

ABSTRACT

This note collects the falsifiable predictions of the rope hypothesis in one place, for outside review. The hypothesis is a non-standard, self-published research programme that attempts to give physics a single physical object — a two-strand rope under tension — from which the standard phenomena follow. Its speculative status is stated plainly throughout; the purpose here is not to argue the framework but to expose its empirical commitments to falsification. Four predictions are testable by experiments already underway or planned this decade: (1) the MOND acceleration scale $g^\dagger = cH_0/2\pi$; (2) the sum of neutrino masses $\Sigma m_\nu = 58.5$ meV; (3) a cosmic-birefringence cross-correlation $EB/EE = 0.0119$, flat in multipole; and (4) a specific MOND interpolation-function shape differing $\sim 8\%$ from standard forms near the transition. Each is stated with its number, its test, the experiment that will deliver the verdict, and the conditions that would falsify it — and, critically, with an explicit ledger of which inputs are derived from the framework versus taken from measurement. Every number is reproducible from an open, tested software package (`rope_solver`), so a reviewer can recompute each prediction rather than trust it.

This supersedes the earlier note “Two Falsifiable Predictions” (June 2026), adding two predictions, a structural fifth, and the reproducibility standard.

1. How to Read This Note

A prediction is only as good as its honesty about what stands behind it. For each prediction below we separate three things a reviewer should never have to untangle on their own:

- The number — the specific quantity the framework commits to, with no remaining freedom to adjust.
- The inputs — stated explicitly as DERIVED (forced by the rope structure) or MEASURED (taken from experiment as a fixed input). A prediction that leans on a measured input is still falsifiable, but it is not parameter-free, and we say so.
- The test and the falsifier — the experiment that will decide, and the result that would kill the prediction.

None of these predictions depends on accepting the rope ontology. Each follows from a stated mathematical relation; a reader may accept, test, or reject any one independently of the others and of the framework that motivated it.

REPRODUCIBILITY

Every number in this note is computed by an installed function in the open rope_solver package, with a regression test pinning it. A reviewer can run three commands (validation 10/10, physics+EM 59/59, reproduction 6/6) and recompute each prediction from source. The framework also publishes, in the same package, an explicit registry of what it has NOT derived and what it has ruled out — including a failed mechanism for particle masses. Predictions and limitations are held to one standard. Each prediction maps to a frozen benchmark ID (e.g. EM-005 for the birefringence number), so a reviewer can cite and re-run any single result.

2. The Predictions at a Glance

#	Prediction	Number	Test (experiment)	Verdict by
1	MOND scale $g^\dagger = cH_0/2\pi$	$1.04 \times 10^{-10} \text{ m/s}^2$	H_0 convergence; z-drift of $g^\dagger$	DESI/Euclid, ongoing
2	Neutrino mass sum Σm_ν	58.5 meV	CMB + BAO + galaxy surveys	CMB-S4/Euclid, ~2027+
3	Cosmic birefringence EB/EE (EM-005)	0.0119 (flat in ℓ)	CMB EB cross-power	LiteBIRD, ~2030s
4	MOND interpolation shape	~8% vs standard	Galaxy rotation curves	SPARC, available now
5*	W/Z share lepton Koide phase	structural	(not currently testable)	—

* Predictions 5 and 6 are structural consequences, stated for completeness; they are not counted among the four empirical predictions. Prediction 5 is not currently testable; Prediction 6 (correlated α - G variation) is testable in principle against varying-constants data, but its specific coefficient is provisional on deriving the gravity-sector relation $G(T, \kappa, a)$.

3. Prediction 1 — The MOND Acceleration Scale

The relation

The MOND characteristic acceleration $g_{\dagger} \approx 1.2 \times 10^{-10} \text{ m/s}^2$ is an empirical parameter in Modified Newtonian Dynamics and in the Radial Acceleration Relation observed in disk galaxies. The rope framework proposes it is not free but fixed by the Hubble rate:

$$g_{\dagger} = c H_0 / 2\pi = c^2 / (2\pi R_H)$$

connecting the galactic acceleration scale directly to cosmology using only c , H_0 , and a geometric factor 2π . No galaxy-specific fitting enters.

Inputs (derived vs measured)

- **MEASURED:** H_0 (the Hubble constant). The prediction inherits the Hubble tension — an irreducible $\sim 9\%$ spread ($H_0 = 67\text{--}73 \text{ km/s/Mpc}$) until H_0 is pinned.
- **DERIVED (partially):** the form $cH_0/2\pi$ from the horizon coupling. **HONEST CAVEAT:** the 2π is motivated by horizon-sphere geometry but is not uniquely forced; the alternative $cH_0/6$ performs comparably against SPARC (RMS 0.203 vs 0.201). We do not claim the coefficient is settled.

Test and falsifier

With current local H_0 , $g_{\dagger} = cH_0/2\pi$ sits about 6% below the SPARC value — right ballpark, not confirmed. The sharper, framework-specific test is drift: $g_{\dagger}$ and H_0 should track together across cosmic time. A measurement of $g_{\dagger}$ from high-redshift galaxy samples that does NOT scale with $H(z)$ would falsify the relation. The prediction is also falsified if H_0 converges far from the value that makes the relation hold, with no z -drift compensating.

4. Prediction 2 — The Sum of Neutrino Masses

The relation

The Koide formula relates the three charged-lepton mass amplitudes to a phase on a circle; the measured lepton masses fix that phase at $\varphi = 132.733^\circ$. Applying the same Koide structure to neutrinos with a fixed offset $\pi/12 = 15^\circ$ (Brannen 2006) gives $\varphi_\nu = 147.733^\circ$. With the measured solar splitting as the one scale input, all three neutrino masses are then determined.

$$\Sigma m\nu = 58.5 \text{ meV}$$

Inputs (derived vs measured)

- MEASURED: the solar mass splitting Δm^2_{21} (the one scale), and the ratio $\Delta m^2_{31}/\Delta m^2_{21} = 32.6$ used to fix the offset.
- INPUT, NOT DERIVED: the $\pi/12$ offset itself. HONEST CAVEAT: $\pi/12$ is an empirical choice, not derived from the rope structure. A skeptic who treats the offset as fitted will regard $\Sigma m\nu$ as a consistency check rather than a pure prediction. We state it as the former.

Test and falsifier

$\Sigma m\nu = 58.5 \text{ meV}$ is the one number here not yet measured, with nothing left to adjust. It sits just above the normal-ordering minimum ($\sim 59 \text{ meV}$), making it sharp and risky: it is distinguishable both from the quasi-degenerate spectrum ($\Sigma \sim 100\text{--}200 \text{ meV}$) and, with care, from the bare splitting minimum. The prediction is falsified by a confirmed $\Sigma m\nu$ below $\sim 50 \text{ meV}$ or above $\sim 70 \text{ meV}$, or by confirmation of inverted ordering. Cosmological bounds are already pressing toward this range.

5. Prediction 3 — Cosmic Birefringence (EB/EE)

The relation

The rope is a chiral object: its two strands wind with a definite handedness. That chirality acts on propagating light as a Chern-Simons-type coupling, rotating the plane of linear polarization as it travels — cosmic birefringence. Integrated to the last-scattering surface, the rotation angle is

$$\beta = \chi c D / 2, \quad \chi = \sin(2\theta_W) n_{\text{rope}} r_H^2 / c$$

Eskilt & Komatsu (2022) report $\beta = 0.342^\circ \pm 0.094^\circ$ from Planck. Matching that fixes the rope chirality product to $\sin(2\theta_W) n_{\text{rope}} r_H^2 = 9.18 \times 10^{-29} \text{ m}^{-1}$. The framework then predicts the CMB EB cross-correlation:

$$EB/EE = \sin(4\beta)/2 = 0.0119, \quad \text{flat in multipole } \ell$$

Inputs (derived vs measured)

- DERIVED: the mechanism (a parity-odd CS coupling from the rope helix) and the functional form $EB/EE = \sin(4\beta)/2$, flat in ℓ . The flatness is a genuine rope-specific signature — many alternative mechanisms predict ℓ -dependence.
- MEASURED: the rotation angle β (from Planck), which fixes the chirality product. The individual geometric factors (θ_W , n_{rope} , r_H) are not separately determined — only their product is constrained.

Test and falsifier

LiteBIRD (launch ~2030s) is designed to measure CMB EB to the precision this prediction requires. The prediction is falsified if EB/EE is found inconsistent with $\sin(4\beta)/2$ at the measured β , or if the cross-correlation shows significant multipole dependence rather than the predicted flatness. This is the cleanest near-term test of a rope-specific mechanism: an outside experiment, a sharp number, and a distinctive spectral shape.

6. Prediction 4 — The MOND Interpolation Shape

The relation

Beyond the acceleration scale (Prediction 1), the rope tension response in the weak-field regime fixes the SHAPE of the transition between Newtonian and modified dynamics — the interpolation function $v(y)$. The rope form is a tanh-type function, distinct from the “simple” and “standard” interpolation functions commonly fit to data. The forms differ by about 8% in the transition region (accelerations near $g^\dagger$), where rotation-curve data are most constraining.

Inputs (derived vs measured)

- DERIVED: the interpolation shape from the rope tension’s nonlinear response. No per-galaxy fitting of the shape is permitted — it is a single fixed function for all galaxies.
- MEASURED: only the overall scale $g^\dagger$ (Prediction 1) enters; the shape adds no new free parameter.

Test and falsifier

The SPARC database of high-quality rotation curves is available now and resolves the transition regime for many galaxies. The prediction is falsified if the data systematically prefer a different interpolation shape (e.g. the standard $y/\sqrt{1+y^2}$ form) over the rope tanh form at a significance the ~8% difference makes accessible. This is the one prediction testable with existing data today, and therefore the natural first target for a skeptical reviewer.

7. Prediction 5 (Structural) — Gauge-Boson Koide Structure

At the rope's Chern-Simons level $k = 3$, the quantum dimensions of the spin- $\frac{1}{2}$ and spin-1 representations coincide ($d_{\frac{1}{2}} = d_1 = \Phi$, the golden ratio, because $\sin(2\pi/5) = \sin(3\pi/5)$). This forces the W and Z bosons onto the SAME phase structure $\phi = (3+\Phi)\theta_W$ that governs the charged leptons — a lepton/gauge-boson link the Standard Model does not contain. The underlying identity ($d_{\frac{1}{2}} = \Phi$ at $k = 3$) is exact mathematics. We state this as a structural prediction only: it is a clean consequence of the framework but we do not have a direct experimental test for it, and we do not count it among the four empirical predictions above.

8. Prediction 6 (Structural / Provisional) — Correlated Variation of α and G

The recent microscopic work expresses the electromagnetic coupling and the gravitational coupling in the SAME rope-medium primitives. The electromagnetic energy coefficient is $c = \kappa a / (3T^2)$, giving a dimensionless EM strength $\alpha \sim 2T^2/(\kappa a)$; the gravitational coupling is governed by the same rope tension T . Because both couplings are functions of one shared set of primitives (tension T , shared-atom bond stiffness κ , rope spacing a), α and G are NOT independent — a point on which this framework differs sharply from standard physics, where they are unrelated.

The relation

If a shared primitive drifts (in space or time), α and G must co-vary in a fixed, scale-free ratio. For drift in the shared tension T , differentiating $\alpha \sim 2T^2/(\kappa a)$ and the natural rope gravity-rigidity scaling $G \sim 1/(T a)$ gives:

$$d \ln \alpha = -2 d \ln G \quad (\text{tension-driven drift})$$

equivalently $\dot{\alpha}/\alpha = -2 \dot{G}/G$. The coefficient is a pure number, independent of the undetermined absolute rope scale — which is why this survives as a prediction even though the individual primitives are not yet independently fixed. It also differs from generic varying-constants models, which impose no fixed relation between α and $\dot{G}$.

Inputs (derived vs assumed)

- $\alpha \sim 2T^2/(\kappa a)$: from the EM energy coefficient $c = \kappa a / (2T^2)$, whose chain (boundary continuity $\rightarrow$ energy minimization $\rightarrow$ EFT form $\rightarrow$ microscopic coefficient) is derived to rope-medium primitives, with the locking energy $J = T^2/\kappa$ now EXACT from endpoint mechanics in the harmonic regime.
- $G \sim 1/(T a)$: the natural tension-rigidity scaling of the gravity sector. This is an ASSUMED scaling, NOT yet derived to the same rigor as the EM coefficient. The specific exponent -2 depends on it.

Test and falsifier

Testable now against combined bounds: quasar-absorption limits on α and lunar-laser-ranging / pulsar-timing limits on $\dot{G}$. If a tension-driven drift of the constants is ever detected, the framework requires $\dot{\alpha}/\alpha = -2 \dot{G}/G$; a measured ratio inconsistent with -2 (for tension-channel drift) falsifies the shared-tension structure. Two honest qualifications: (i) the number -2 is PROVISIONAL on deriving $G(T, \kappa, a)$ rigorously — the robust, framework-forced content is that α and G co-vary in a FIXED ratio when a shared primitive drifts, not the specific value; (ii) current data are consistent with NO variation of either constant, so this is at

present a bound and a standing prediction, not a detection. Deriving $G(T, \kappa, a)$ to the EM sector's standard is the single step that would turn the ratio into a sharp number the framework stakes itself on.

9. Honest Status of the Framework

For any reviewer evaluating these predictions, the following should be stated plainly:

- The rope hypothesis is non-standard, self-published, and not peer-reviewed. These predictions are presented as empirical consequences of specific stated relations, not as derivations from established physics.
- Two of the four predictions (the neutrino $\pi/12$ offset; the MOND 2π coefficient) rest on inputs that are empirical choices, not derived. They are falsifiable but not parameter-free, and we have marked exactly where.
- The framework's deepest open problem — deriving absolute particle masses (and hence atomic and molecular masses) — is **UNSOLVED**. A systematic search ruled out eight natural mechanisms; dimensional transmutation was closed by a structural result (the rope's dimensionless sector sits at a conformal fixed point, so it cannot run to generate a mass hierarchy). The framework openly publishes this failure alongside its successes.
- The strongest internal result — lepton mass RATIOS from $\varphi = (3+\Phi)\theta_W$ to $\sim 0.5\%$ — is a **CONJECTURE**, not a theorem: its load-bearing step (a time-reversal parity assignment) rests on the nonzero chiral central charge $c = 9/5$, which is rigorous, but two further assumptions remain, and the result requires the measured Weinberg angle. It is not a parameter-free mass prediction and is not listed among the four above.

THE INVITATION

These four predictions are offered for falsification. The framework that motivates them is speculative and incomplete, and its incompleteness is documented as carefully as its claims. A reviewer does not need to engage the ontology to test the numbers. If the cleanest test (Prediction 4, available now in SPARC) or the sharpest (Prediction 3, LiteBIRD) fails, the corresponding relation is wrong, and we want to know.

Acknowledgment of Origin

The core idea underlying this work — that physical phenomena arise from a two-strand rope-like structure connecting matter, with light as a transverse disturbance along the rope, and the principle that physical theories must describe objects rather than non-object concepts — originates with Bill Gaede (the “Rope Hypothesis” / “Thread Theory”). The predictions and the computational, field-theoretic, and topological development presented here are an independent, modified mathematization that departs from his formulation in substantive ways (for example, treating magnetism as a vortex rather than a jump-rope

structure). Crediting this origin is not a claim that Gaede endorses this work, nor an endorsement of all of his positions; it is the honest acknowledgment of where the foundational idea came from.

References

- Eskilt, J. R. & Komatsu, E. 2022, Phys. Rev. D 106, 063503 (cosmic birefringence).
Brannen, C. 2006, unpublished (Koide neutrino phase offset $\pi/12$).
Koide, Y. 1983, Phys. Lett. B 120, 161 (charged-lepton mass relation).
Lelli, F., McGaugh, S. & Schombert, J. 2016, AJ 152, 157 (SPARC database).
Software: rope_solver 2.0 — open package, frozen APIs and benchmark IDs; every prediction reproducible from source with regression tests and an open registry of unsolved/falsified items.

Correspondence: Mark Palmer, Global Networking Practice, NTT DATA, Charlotte NC. Physics correspondence to palmer100@gmail.com. The author writes in a personal capacity. Numerical values computed with rope_solver 2.0 (open source); a reviewer can reproduce every figure in this note from the package.

Two Predictions from Strand Extensibility

1. Vacuum optical nonlinearity (Kerr-type). The strand stretch modulus k , required for the repulsive core (EM-RECON-009), makes light self-interact at quartic order with onset strain $g^* = \sqrt{4T_0/(k-T_0)}$ (EM-RECON-010). Identification with measured vacuum nonlinearity (light-by-light scattering; vacuum birefringence) measures k from pure optics; a functional form differing from Euler–Heisenberg would discriminate the rope mechanism.
2. A dark, gapless, faster-than- c longitudinal tension channel. Stability requires $k > T_0$, so longitudinal strand waves propagate at $c_{L,eff} = \sqrt{((k+\lambda\gamma^2\tau_0^2)/\mu)} > c$, with $\gamma = 1/\sin^2\theta$ derived from helix geometry (EM-RECON-012). The channel decouples from matter and light at linear order exactly (cubic-only coupling) and creates no causal paradox in the emergent-Lorentz framework (EM-RECON-011). Any observed linear coupling to a superluminal stress channel, or its firm exclusion at the predicted nonlinear coupling strength, bears directly on the extensibility primitive.

The Vacuum-Nonlinearity Prediction, Quantified and Confronted

The polarization structure of the rope vacuum birefringence is now computed exactly: anisotropy ratio 3:1 with NEGATIVE index shifts (background fields stiffen the network; light speeds up), versus QED's 7:4 with positive shifts — a double qualitative discriminator (EM-RECON-016). Confronting the magnitude $|\Delta n| = \frac{1}{2}(k/T_0-1)\epsilon_0 c^2 B^2/\Sigma$ with the final PVLAS limit $\Delta n = (12\pm 17)\times 10^{-23}$ at 2.5 T EXCLUDES the identification of the rope quartic with the observed light-by-light nonlinearity ($\Sigma \approx 1.5\times 10^{25}$ J/m³ overshoots by $\sim 570\times$) and sets $\Sigma \gtrsim 8.6\times 10^{27}(k/T_0-1)$ J/m³. Decisive window: at this bound the rope $|\Delta n|$ still exceeds QED's prediction with opposite sign, so vacuum-birefringence experiments improving toward QED sensitivity will either detect a negative Δn with 3:1 anisotropy (rope signature) or push Σ higher and confirm QED. Either outcome is informative; the framework has already been corrected once by this confrontation.

The Heavy-Hydride 90-Degree Asymptote

The molecular-geometry theorems make 90 degrees the raw fixed point for single bonds on orthogonal dipolar lobes, approached from above as the positive opening corrections collapse down each column. Confirmed on eight hydrides (H_2O 104.48° $\rightarrow$ H_2Te 90.25°; NH_3 106.67° $\rightarrow$ SbH_3 91.70°): monotone, one-sided (no member below 90° — a single 89-degree hydride would have falsified the structure), with the $XH_3 > XH_2$ crowding law holding at every period and the opening's decay length matching the hydrogen core width used independently by two earlier sessions. STANDING PREDICTIONS for members not yet well measured: H_2Po between 90.0° and 90.7°; BiH_3 between 90.0° and 91.5° — one-sided windows, registered in advance. Tier label: consistency-tier against quantum chemistry (which

predicts these angles too); the distinctive content is the asymptote's derivation from two integration-verified theorems rather than a hybridization narrative. Benchmark:

benchmarks/em/heavy_hydride_asymptote.py.

The Measurement-Sector Predictions (added at v2.2.5, July 2026)

With the v2.2.5 cycle, the measurement chain — amplitude, detection event, reservoir, and reservoir spectrum — was constructed on literal strands (FND-STRAND-005 through 008), and a new family of falsifiable commitments follows, stated in the same discipline as the predictions above: what is committed, what is derived versus assumed, and what result would kill each. One structural caveat governs the family: the framework's standing absence of an absolute scale means several of these commit to SHAPES and RELATIONS while leaving locations in frequency or length open.

Prediction 7 — The Gapped Environmental Noise Floor (structural; scale-open)

The relation: the weave reservoir's normal-mode spectrum is gapped — every mode carries the strand mass, band $[\omega_{\text{gap}}, \omega_{\text{gap}} \sqrt{1 + 4 \kappa t}]$, NO soft modes; environmental noise sourced by the fundamental medium arrives only at and above the gap. DERIVED: the gap's existence and the band shape, from the engine's own Hessian (FND-STRAND-008; matched to dispersion at 7e-16). OPEN: the gap's location in hertz. Falsifier: an established, irreducible, gapless environmental noise spectrum extending to zero frequency at every accessible scale, attributable to the fundamental medium rather than to ordinary matter environments.

Prediction 8 — Detection Is Nucleation Kinetics

The relation: the detector click is a thermally activated topological event (kink-pair nucleation, FND-STRAND-006/007), committing to the full first-passage package where the model applies: near-exponential waiting times at fixed drive, Arrhenius dark counts, latency decreasing with channel energy. DERIVED: the event's existence, localization, irreversibility, and channel-energy drive (Arrhenius-linear in amplitude-squared, r-squared 0.984). HONEST CAVEAT: parts of this package resemble known avalanche-detector phenomenology — the overlap is why the model is plausible, and the commitment is to the WHOLE package. Falsifier: confirmed click statistics that are non-activated and non-first-passage in a regime where the model's detector physics applies.

Prediction 9 — Spin-1/2 Requires Extension (in-principle)

The relation: the half-angle response is the SU(2) lift, which a point orientation cannot carry and a framed strand cannot avoid (FND-STRAND-005; measured at 1e-9 for arbitrary axes and profiles). Commitment: every spin-1/2 particle has internal structure at SOME finite scale. Scale-open, hence one-sided in practice: compositeness bounds may rise indefinitely without contradiction, but a CONFIRMED structureless point with spin-1/2 at all scales would falsify the mechanism outright.

Prediction 10 — Tsirelson Forever, with the Failed Organ Named

The relation: CHSH-type correlations never exceed $2 \sqrt{2}$ — shared with quantum mechanics; distinctive here is the WHY and therefore the autopsy: supra-Tsirelson statistics require a setting-dependence channel of polynomial degree greater than 24 where this framework's physics has degree exactly 1 (QB-018/019/022). Falsifier: any confirmed loophole-free violation above $2 \sqrt{2}$ — which kills the measurement sector at a named claim, not merely embarrasses it.

Prediction 11 — Born's Square from Channel Energy

The relation: the quadratic ($p = 2$) response near detection threshold is the statement that nucleation is driven by channel ENERGY, quadratic in amplitude (QB-011 protocol; FND-STRAND-005 B4, $p = 1.99$). Falsifier: a confirmed, reproducible deviation from quadratic threshold response — breaking the chain at a specific link (the energy-drive identification), leaving the rest standing for inspection.

Prediction 12 — Analog Universality (testable now)

The relation: any engineered system realizing the twist-chain physics — coupled pendulum arrays, magnetic domain-wall lattices, cold-atom sine-Gordon simulators — must exhibit the friction cliff (seven-order exponential mobility suppression, slope approximately -8.5 per width unit, arrest/coast dichotomy) with the LAW transferring across implementations and only prefactors carrying the coupling convention (FND-KIN-002; FND-STRAND-002, slope agreement 0.16 percent). Engineered filament brushes should likewise show the transparent-to-solid coverage crossover at gap approximately the contact size with the

measured inverse-fourth-power approach (FND-STRAND-004). No knowledge of the absolute strand scale is required, because the experimenter builds the system; these are testable with current laboratory technique.

A retrodiction, declared as such: the Calugareanu ledger and the supercoiling instability (FND-STRAND-003) are confirmed daily by DNA mechanics — consistency evidence for the strand engine, counted here as retrodiction, not prediction.

Registered Claims in This Paper

Each statement below is traceable to the machine-readable registry (claims.yaml), the corpus's single source of truth. Status labels: Derived (follows from stated mechanics), Modeled (reproduces data with declared inputs), EFT-constrained (bounded by effective-theory limits), Conjecture (proposed, not derived), Open (registered unsolved problem), Failed (falsified, kept as a finding). Where a claim is code-backed, the named benchmark reruns it.

- **CHEM-GEO-002 [Derived]:** HEAVY-HYDRIDE 90-DEGREE ASYMPTOTE [benchmark: benchmarks/em/heavy_hydride_asymptote.py]
- FND-STRAND-005 [Modeled]: THE INTERACTION MODEL ON LITERAL STRANDS — the framed strand is the SU(2) lift [benchmark: benchmarks/foundations/strand_interaction_model.py]
- FND-STRAND-006 [Modeled]: THE NUCLEATION EVENT ON STRANDS — the dot as topological event [benchmark: benchmarks/foundations/strand_nucleation.py]
- FND-STRAND-007 [Modeled]: THE WEAVE AS RESERVOIR — the bath derived [benchmark: benchmarks/foundations/strand_weave_reservoir.py]
- FND-STRAND-008 [Modeled]: THE WEAVE SPECTRUM FROM THE ENGINE — the bath is gapped [benchmark: benchmarks/foundations/strand_weave_spectrum.py]
- QB-019 [Derived]: MECHANISM-UNITY SELECTS TSIRELSON [benchmark: benchmarks/bell/tsirelson_selection.py]
- QB-022 [Modeled]: MECHANISM-UNITY DECOMPOSED AND GROUNDED (SOVO) [benchmark: benchmarks/bell/mechanism_unity_grounded.py]